PRODUCT DOSSIER · VERSION 3.0 · NOVEMBER 2026

AIX ALPHA v3.0

The loop-engine AI workspace: chat, Studio, connectors, credits and members, in one complete product.

4	8	6	$0.10
ROUTING TIERS	MODEL PERSONAS	CORE TOOLS	PER CREDIT

Prepared by the BONSAI Division of AIX · Confidential product overview

Figures in this document are generated from the live app configuration.

01 · WHAT AIX ALPHA IS

Overview

AIX Alpha v3.0 is a complete AI workspace built around a loop engine: a set of deterministic protocols that decide how little context, how few tokens and how few model calls each request really needs. Everything the user touches (chat, Studio, tools, connectors, members and analytics) sits on top of that engine, and everything is metered in one unit: credits.

0.0014	$0.10	3	1
SPEND PER CREDIT UNIT	CUSTOMER PRICE / CREDIT	QUALITY MODES	SINGLE-FILE BUILD

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01	Overview	Product map and principles
02	The Loop Engine	Eight layers that save tokens
03	Models and personas	Eight engines, three modes
04	Credits and pricing	Packs, per-model cost, estimator
05	Studio	Per-session app builder
06	Chat	Toolbar, notes, attachments, voice
07	Tools, MCP and connectors	Plug-and-play inside replies
08	Members, users and analytics	Database, balances, Stripe plan
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Product map

CHAT
Eight personas, notes, attachments, voice input, per-message toolbar and connectors that answer inside replies.

STUDIO
Session-scoped app builder with live preview, minimal-edit patches, version history, visual editing and a smart debugger.

TOOLS AND CONNECT
Six core tools, a plugin registry and plug-and-play MCP connectors (PayPal, Gmail, Claude Docs).

CREDITS
One metered unit across every feature, with per-mode estimates shown before you run anything.

MEMBERS
Pay-as-you-go packs, balances, a users and tokens database with a users.md export and Stripe-ready links.

SITES AND ANALYTICS
Ready presets for common builds and usage analytics broken down by model and mode.

02 · WHY REQUESTS COST LESS

The Loop Engine

Every request passes through the same pipeline. Most steps are plain code that costs nothing; the model is called only when it is needed, with the smallest prompt that can still do the job.



Layer	What it does	Saves
Reflex	Answers greetings, simple math and fixed commands with no model call.	Whole request
Router	Classifies each request into NANO, MICRO, STANDARD or DEEP and sets token budgets and thinking depth.	Over-provisioning
Context diet	Sends a compact prompt core, code stubs instead of full files, a rolling digest, notes and a stepped history window.	Input tokens
Response cache	Identical requests reuse a stored answer within the session.	Repeat calls
Generate	One bounded call with a strict output cap per tier.	Output tokens
Verify	Deterministic checks: truncation, broken code fences, HTML or script tag balance, valid JSON, word and bullet limits, repetition.	Judge calls
Continue / refine	If output was cut off the engine appends only the missing tail; if a check fails it repairs just that part.	Full retries
Judge (optional)	A low-cost LLM review used only in Thorough mode.	Wasted quality spend
Ledger and caps	Every call is converted to credits and tested against per-chat, per-day and per-request caps.	Runaway spend

Routing tiers

Tier	Typical request	Typical in / out tokens
NANO	Greeting, yes/no, lookup handled by reflex or minimal prompt	Under 100 / under 60
MICRO	Short answers, small edits, quick questions	260 / 120
STANDARD	Normal chat, explanations, medium builds	520 / 450
DEEP	Reasoning, multi-file apps, audits, long writing	900 / 2600

Minimal-edit rule. When changing existing work, AIX Alpha produces SEARCH/REPLACE patches rather than rewriting whole files. A full rewrite only happens when the user asks for one, which keeps output tokens, and credits, low.

03 · EIGHT ENGINES, ONE WORKSPACE

Models and personas

All personas run on the same efficient engine. They differ in prompt core size, thinking depth and output style, which is what moves the credit cost. Prompt cores are measured in tokens.

Persona	Best for	Thinking	Style	Core tok
AIX-ALPHA-V3	Everyday answers, apps and edits at the lowest cost	LOW	Concise	138
AIX-ALPHA-QUANTUM-V1.0	Hard multi-step reasoning and decisions under uncertainty	HIGH	Detailed	143
AIX-ALPHA-LE-V1.0	Quick facts and short answers with zero filler	LOW	Very concise	56
AIX Code Architect	Full-stack code and multi-file apps	HIGH	Detailed (code)	64
Executive Strategist	Decision memos, risks and trade-offs	LOW	Standard	45
STEM Tutor	Step-by-step explanations and learning	LOW	Detailed	41
Security Auditor	Code and architecture audits	HIGH	Detailed	56
Creative Storyteller	Creative writing, hooks and copy	LOW	Detailed	30

Quality modes

Mode	Refine passes	Output scale	Cap scale	Use when
Eco	1	0.85x	0.85x	Fast, lowest credit use
Balanced	2	1.0x	1.0x	Default for daily work
Thorough	3	1.3x	1.5x	Higher stakes; adds judge pass

Thinking level

LOW personas keep reasoning short and answer directly. HIGH personas (Quantum, Code Architect, Security Auditor) reserve a larger reasoning budget on DEEP requests, which is why their DEEP cost is higher in the next section.

04 · PAY AS YOU GO, NO TIES

Credits and pricing

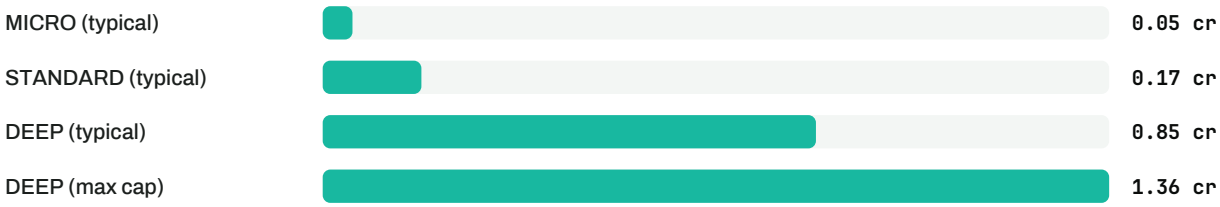
One credit costs $0.10. Credits are bought in packs, never as a subscription, and every feature draws from the same balance, including Studio.

$10	$5	$100	0
100 CREDITS · STARTER	50 CREDITS · TOP-UP	1,000 CREDITS · ENTERPRISE	MONTHLY OR YEARLY TIES

Pack	Price	Credits	Per credit	Notes
Starter	$10	100	$0.10	One-time buy-in that opens your account.
Pay as you go	$5	50	$0.10	Top up in 50-credit packs whenever you run low.
Enterprise	$100	1,000	$0.10	Bulk 1,000-credit pack with no monthly or yearly commitment.

Cost per request by tier

Credits per single request on the default engine in Balanced mode, typical versus maximum (cap), with thinking off. Dollar values use $0.10 per credit.



Per-model cost, Balanced mode (credits per request, typical)

Model	MICRO	STANDARD	DEEP	DEEP + thinking
AIX-ALPHA-V3	0.051	0.170	0.85	0.93
AIX-ALPHA-QUANTUM-V1.0	0.065	0.219	1.13	1.21
AIX-ALPHA-LE-V1.0	0.039	0.140	0.70	0.78
AIX Code Architect	0.061	0.221	1.17	1.25
Executive Strategist	0.051	0.188	0.98	1.06
STEM Tutor	0.057	0.211	1.12	1.20
Security Auditor	0.058	0.212	1.12	1.20
Creative Storyteller	0.059	0.219	1.17	1.25

Cost by mode (default engine)

Mode multiplies output length and the number of refine passes. Each figure is credits per typical request at that tier.

Mode	MICRO	STANDARD	DEEP	Thinking on · DEEP
Eco	0.046	0.150	0.73	0.81
Balanced	0.051	0.170	0.85	0.93
Thorough	0.062	0.211	1.09	1.17

What a pack really buys

Average credits per request depend on how a person works. The spend estimator in the About page uses these mixes of MICRO, STANDARD and DEEP requests on the default engine.

Usage profile	Mix (M / S / D)	Eco	Balanced	Thorough	Requests per $10 (Balanced)
Light chat	70 / 25 / 5	0.11	0.12	0.15	827
Balanced	50 / 35 / 15	0.18	0.21	0.27	470
Builder	20 / 40 / 40	0.36	0.42	0.53	239

No surprises. The credit estimate for the selected persona, mode and thinking setting is shown before a request runs, and again as a live meter while Studio builds. Caps stop any single request, chat or day from exceeding its limit.

Where credits are spent

CHAT REPLIES

Billed by the tier the router picked, plus any refine or continue pass.

STUDIO BUILDS AND EDITS

Generation, patches, debugger runs and visual edits all draw from the same balance and are logged in the Studio activity timeline.

TOOL AND CONNECTOR CALLS

Model time to choose and summarise a tool call. External services bill separately under their own accounts.

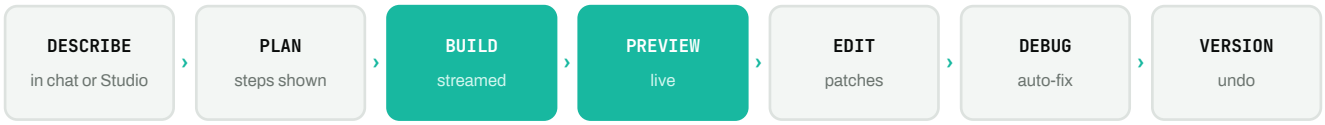
FREE ACTIONS

Reflex answers, cache hits, previews, history restore, exports and the users.md download use no credits.

05 · BUILD, PREVIEW, DEBUG, ITERATE

Studio

Studio is a full app builder inside the workspace. Each chat session owns its own projects, files and Studio conversation, so nothing leaks between sessions, and the Studio chat on the right keeps the complete activity trail.



PER-SESSION APPS AND CHATS

Apps and the Studio conversation are saved with the session. Start a new session and you get a blank project; return to an old one and everything is as you left it.

GENERATE FROM CHAT

Code in a chat reply offers Edit in Studio or Preview in Studio when you are in that session, so ideas move straight into a build.

LIVE BUILD LOG

Step-by-step activity like a modern agent: what it is doing, which file, what changed, and a streaming preview while it works.

MINIMAL EDITS

Changes arrive as SEARCH/REPLACE patches. Only the touched lines change; the rest of the file is never regenerated unless you ask.

VERSION HISTORY AND UNDO

Every change is snapshotted. Step back to any earlier preview or restore it without spending credits.

VISUAL EDIT MODE

Click an element in the preview and describe the change. The selection is sent as precise context so edits stay small.

DEVICE MODES

Desktop, tablet and phone frames for checking responsive behaviour before you ship.

LINT AND STATIC CHECKS

Catches unclosed tags, broken scripts and common mistakes before the preview even loads.

Smart debugger

When a preview misbehaves, the debugger reads the captured console output, runs static checks, and proposes a minimal patch. Previews that call outside services are automatically given a fetch shim with realistic sample API data, so a weather app, a store or a dashboard renders with real-looking content instead of an empty page even inside the sandboxed preview.

Step	What happens	Cost
Capture	Console errors, warnings and failed requests are collected from the preview.	Free
Diagnose	Static lint plus the captured errors are summarised into a short brief.	Free
Patch	The model returns a SEARCH/REPLACE patch for only the failing code.	MICRO or STANDARD
Verify	Preview reloads; if errors remain the loop repeats within the mode's refine limit.	Per pass

05 · READY TO START

Studio presets

The Sites area ships with presets that load a complete multi-file project into Studio in one click.

Preset	Description	Files
Multi-Page Web & TSX App	Linked multi-page application with navigation router, numbers.html sub-page, and interactive React TSX component	5
Quantum SaaS Web Platform	Ultra-modern monochromatic SaaS application with Tailwind CSS, animated metrics, interactive pricing, and dark interface	4
Minimalist Portfolio & Studio	Clean Apple-style portfolio showcase with responsive bento gallery, project drawer, and contact form	2
HTML5 Canvas Retro Arcade	Full 60fps HTML5 Canvas space dodge game with physics loop, keyboard controls, and score tracking	3

Studio and credits

Studio spend is counted against the same balance as chat. A running estimate is displayed in the Studio header, each activity entry shows what it used, and the Progress panel summarises the total for the session.

> Why it stays cheap. Context diet sends stubs of unchanged files, patches replace regeneration, and the preview, lint and history features that make iteration fast are all free of model cost.

06 · EVERYTHING AROUND A REPLY

Chat

Chat is the front door. It supports persona selection, quality modes, thinking toggle and the estimates for each, with a layout that is full width on desktop and fits mobile screens.

MESSAGE TOOLBAR
Hover any message to reveal icons at the top right: copy, edit and resend, add to chat (to keep typing from it), share, and remove.

NOTES
Pinned notes are included in the compact prompt, so context persists without resending whole conversations.

ATTACHMENTS
Add files and images. Large files are reduced to stubs and a digest before any model call.

VOICE
Dictate into the composer using browser speech recognition.

PERSONAS
Switch between eight engines per chat; the credit estimate updates immediately.

SHARE
Share a single response as text or a link-ready block, without exposing the whole session.

Message actions

Action	Behaviour
Copy	Copies the response or message text to the clipboard.
Edit and resend	Loads the message back into the composer; resending branches from that point.
Add to chat	Appends the message to the composer so you can extend or edit before sending.
Share	Opens the share sheet for that one message.
Remove	Deletes the message from the session view and from the context sent to the model, which also reduces future cost.

07 · PLUG AND PLAY

Tools, MCP and connectors

Six core tools are available to every persona. Beyond those, a plugin registry and MCP connectors extend what a reply can do.

Tool	Purpose
aix_quantum_reason	Structured deep reasoning for hard multi-step problems.
search_live_web	Looks up current information from the internet.
generate_web_app	Creates a full web app project and opens it in Studio.
execute_sandbox_code	Runs code in an isolated sandbox and returns the output.
manage_workspace_file	Reads, writes and organises workspace files.
store_memory_kv	Stores and recalls small facts across a session.

MCP connectors

The Connect hub lists connectors with a single toggle. Once connected, you can simply ask in chat and the reply uses the connector and answers inline. Each connector shows example prompts, and write actions are clearly marked.

Connector	Read	Write	Example prompt
PayPal	Check payments, invoices and disputes.	None (read-only)	Show my latest PayPal transactions
Gmail	Search mail, read threads, draft replies.	create_draft	Find emails from the last week about invoices
Claude Docs	Save work as a brand-new document.	None (read-only)	Save this chat as a document

Plugin registry

Plugin	Category	Version	Provides
GitHub Workspace Sync	Developer Tools	1.4.2	github_list_repos, github_create_pr, github_commit_files
PostgreSQL & Vector DB	Database & Storage	2.1.0	db_query, db_vector_search, db_get_schema
Figma Design Token Bridge	Design Systems	1.0.8	figma_extract_tokens, figma_render_component
Slack & Webhook Gateway	Communication	1.2.0	slack_post_message, slack_upload_snippet
Brave Search & Crawler	Search & Crawling	1.1.5	brave_web_search, brave_fetch_dom

08 · ACCOUNTS, BALANCES, BILLING

Members, users and analytics

The Members page is where credits are bought and balances are managed. It shows each pack, what is included and a breakdown of how credits are spent per model and mode. Payment links are Stripe-ready, so checkout can be switched on by pasting links into configuration.

PAY AS YOU GO
Starter $10 for 100 credits, top-ups $5 for 50, and Enterprise $100 for 1,000. No monthly or yearly commitment at any size.

USERS AND TOKENS DATABASE
Stores members, their credit balance and total tokens used. Totals are shown in tokens and credits, not prices.

USERS.MD EXPORT
One click downloads a readable markdown file of members and usage, ready for records or review.

STRIPE-READY
Pack buttons carry placeholder payment links. When live links are added, purchases update the balance.

ANALYTICS
Usage by model, mode and day, with credits and tokens; use it to tune caps and choose a default mode.

CAPS
Per-request, per-chat and per-day limits, configurable, so spend can never run away.

Data collected per member

Field	Description
Member	Display name or identifier
Credits	Remaining balance
Tokens	Total input and output tokens used on the platform
Last active	Most recent request time
Plan	Last pack purchased

09 · DESIGNED TO FEEL PROFESSIONAL

Interface and mobile

The interface is branded AIX ALPHA throughout. The logo lives in the left sidebar, navigation tabs are centred and scale to fit across the top, and every page uses the full available width.

LEFT SIDEBAR

Logo, sessions, quick navigation and typography reused on the About page.

CENTRED TOP TABS

Chat, Studio, Tools, Connect, Sites, Analytics, Members and About scale to fit the header.

PROGRESS PANEL

A running view of steps, plan and results for long tasks, similar to modern agent consoles.

ACTIVITY TIMELINE

Every build step appears as it happens, then collapses into a summary.

MOBILE FIRST

Pages reflow for phones; composer, tabs and Studio panels stack neatly at 390 px.

ABOUT PAGE

A professional overview of models, per-model credit tables, the spend estimator and payments.

10 · TWO WAYS TO RUN IT

Architecture and security

	Artifact build	Server build
Format	One HTML file, shared as an artifact	Express API plus React and Vite front-end
Model access	Built-in sampling through the host	Anthropic API using your key
Data	Artifact database for usage and members	Your own storage
Connectors	Host connectors (PayPal, Gmail, Docs)	MCP server with token auth
Best for	Fast sharing and trials	Self-hosting and full control

Server configuration

Setting	Purpose
ANTHROPIC_API_KEY	Model access
AIX_REQUEST_MAX_USD	Hard ceiling per request
AIX_SESSION_MAX_USD / AIX_DAILY_MAX_USD	Per-session and per-day ceilings
AIX_MCP_TOKEN	Bearer token protecting the MCP endpoint
AIX_ENABLE_SANDBOX	Turns code execution on or off
AIX_MOCK	Runs without calling the model, for tests

Security and data handling

SECRETS
Keys stay in environment variables on the server and are never sent to the browser.

SANDBOXED PREVIEWS
Studio previews run in an isolated frame with a restricted network; outside calls use sample data in the artifact build.

SPEND CONTROL
Per-request, per-session and daily caps in credits and USD.

WRITE ACTIONS
Connector actions that change data are marked and require an explicit request.

11 · TERMS USED IN THE PRODUCT

Glossary

Term	Meaning
Credit	The billing unit; one credit costs $0.10 for customers.
Loop engine	The pipeline of protocols that keeps each request small and verified.
Tier	Routing class (NANO, MICRO, STANDARD, DEEP) that sets token budgets.
Context diet	Techniques that shrink the prompt: cores, stubs, digest, notes, windowing.
Refine pass	A targeted repair of a failed check, limited by mode.
Patch	A SEARCH/REPLACE edit that changes only the lines that need to change.
Persona	A named configuration of prompt core, thinking depth and style.
MCP	Model Context Protocol; the standard for connecting tools and services.
Cap	A configured limit on spend per request, chat or day.
Reflex	A rule-based answer that uses no model call.

Summary. AIX Alpha v3.0 combines a token-saving loop engine, eight personas, a full Studio, plug-and-play connectors and a simple pay-as-you-go credit model, from $10 for 100 credits to $100 for 1,000, with no subscriptions.